Nathnael Tesfaw

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Education

Cornell University, College of Engineering Ithaca, NY
B.S. in Computer Science, Minor in Artificial Intelligence (GPA: 3.5) *Aug 2023 – May 2027*
Coursework: Data Structures, Functional Programming, Algorithms, Machine Learning, Deep Learning, Cybersecurity, Distributed Computing Principles, Systems Programming, Systems for Large-Scale ML

Experience

Workday Pleasanton, CA
Software Engineer Intern *May 2026 – Aug 2026*
- Architected a vulnerability-ownership system as the sole engineer, integrating GUAC (an open-source dependency graph) with Workday's internal tools, approved by leadership for fleet-wide rollout.
- Created a full-stack tool to demo the POC to stakeholders, visualizing the dependency graph and separating each package's own vulnerabilities from inherited ones.
- Designed and deployed Build Doctor, a Python agent that auto-repairs failing builds, expected to cut the failure rate from **30% to 10%**.

Rethink Food (Hack4Impact Cornell) Ithaca, NY
Lead Software Developer *Aug 2025 – Dec 2025*
- Led a team of **7** to ship a production web app for a nonprofit fighting food insecurity in New York City.
- Built a Mapbox platform mapping food-donation flows across **10,000+** restaurants and community orgs to recruit new donors.

Carnegie Mellon University Pittsburgh, PA
Software Engineer Intern *May 2025 – Aug 2025*
- Engineered a GPT-4–powered browser extension for safer online behavior, presented to **hundreds** of Pennsylvania state government representatives.
- Added survey-driven personalization for the extension and a GPT-4 scoring pipeline to evaluate it, raising response actionability **50%** across **6,000+** tests.

Projects

Distributed Key-Value Store | *Java* *Sep 2026 – Present*
- Implemented an exactly-once RPC layer over an unreliable network, guaranteeing at-most-once execution.
- Built replication on Paxos, keeping the store consistent and available as long as a majority of servers are up.
- Adding sharding and transactions to scale the store across replica groups.

DevAudit | *TypeScript, Next.js, Node.js, Claude API, PostgreSQL, Docker, AWS* *Jan 2026 – Sep 2026*
- Built a full-stack AI agent that scans codebases for security flaws and uses Claude to triage the findings.
- Wrote **102** automated tests with CI, and validated the agent across **28** real-world repos in 7 languages.
- Debugged and fixed subtle production bugs, including a race condition that was silently misreporting successful audits as failures.

DreamBooth Re-implementation | *Python, PyTorch, Hugging Face Diffusers, LoRA* *Mar 2026 – May 2026*
- Re-implemented DreamBooth (Ruiz et al. 2023) from scratch, reproducing the paper's subject-driven image generation from just **3–5** photos.
- Extended it with **LoRA** fine-tuning beyond the original paper, matching its benchmark results at a fraction of the compute.

Tower of the Tail | *Java, LibGDX* *Jan 2026 – May 2026*
- Developed core gameplay for a downloadable 2D platformer on a **9-person team**: player mechanics, platforms, and item merge/unmerge logic.
- Owned the Architecture Specification, the design doc guiding the team's engineering across milestones.

Technical Skills

Languages: Python, Java, TypeScript, C, Rust, SQL, GraphQL, Bash, HTML/CSS
Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, Flask, PyTorch, NumPy, Pandas
Databases & Cloud: AWS, Google Cloud Platform, PostgreSQL, MongoDB, Supabase
Tools: Docker, Git, GitHub Actions, Nginx, Linux, Vercel